



Community

Codetopia Mentorship Program - 2026

Git & GitHub



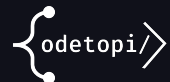
Session 4 :

Undoing Changes



What we will cover today

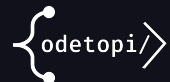
- The Undo Toolkit: restore, revert, reset, amend
- `git restore`: discarding working directory changes
- `git restore --staged`: unstaging files
- `git revert`: safe undo that preserves history
- `git reset --soft` — local-only history rewrite
- Session Project: Undo Drills





Quick Review From Previous Session

- What is HEAD, and what does it normally point to?
- You run `git checkout a3f9c2d`. What state is your repo in now, and how do you safely return to the present?
- What is the difference between `git show <hash>` and `git diff <hash1> <hash2>`?
- In your Session 3 project, you wrote head-explained. Without looking at it, explain detached HEAD state in one sentence.



Undoing Changes



The Undo Toolkit

Four ways to undo in Git

Command	What it undoes
<code>git restore <file></code>	Discard unstaged changes in working directory
<code>git restore --staged <file></code>	Unstage a file (move back from staging area)
<code>git commit --amend</code>	Edit the most recent commit message or contents
<code>git revert <hash></code>	Create a NEW commit that undoes a past commit (safe)
<code>git reset --soft HEAD~1</code>	Undo last commit, keep changes staged (local use only)

git restore: Discarding Changes

Undo an edit you regret

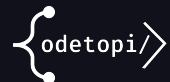
```
# Make a change to a file
echo 'bad idea' >> notes.txt

# Discard working directory changes
git restore notes.txt

# Unstage a file you added by mistake
git add notes.txt
git restore --staged notes.txt
```

!! Warning !!

git restore permanently discards your working directory changes. There is no undo for restore — the changes are gone. Only use it when you are sure you don't need those edits.



git revert: Safe Undo

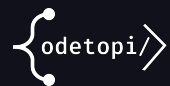
Undo a past commit without rewriting history

```
# Undo the most recent commit  
git revert HEAD
```

```
# Undo a specific past commit  
git revert <commit-hash>
```

What happens:

- Git creates a brand new commit that reverses the changes
- The original commit stays in history — nothing is erased
- Safe to use on shared branches (the team can see what happened)



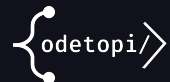
Session Project

Undo Drills

Session Project

Your deliverables

1. Edit a file, then use `git restore <file>` to discard the change
2. Stage a file by accident, then use `git restore --staged <file>` to unstage it
3. Make a bad commit, then use `git revert HEAD` to undo it cleanly
4. Make a messy commit, then use `git reset --soft HEAD~1` to undo and restage
5. Screenshot git log after each operation to document your drill results



References & Further Reading

Resources

Text Resource

- Atlassian: Git Undo Guide

[Undoing Changes in Git | Atlassian Git Tutorial](#)

Video Resource

Web Dev Simplified: Git Reset vs Revert

[Git & GitHub Tutorial for Beginners #7 - Undoing Things](#)

